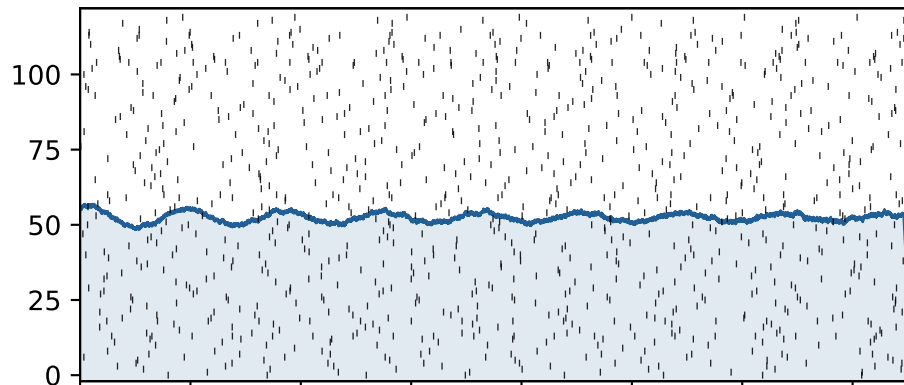
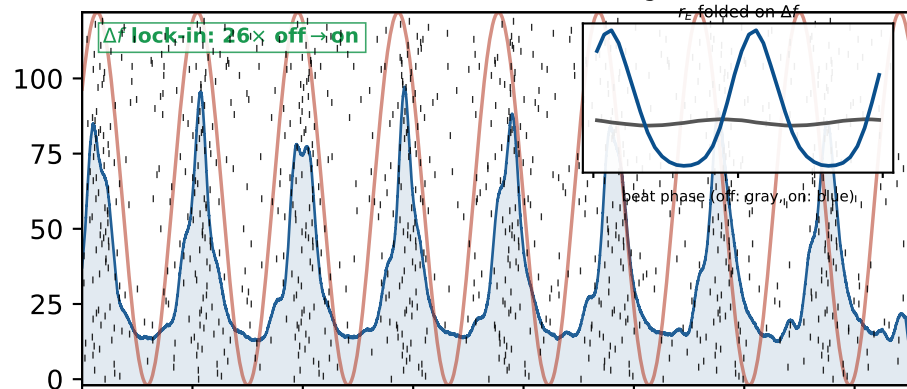


QIF spiking network (microscale of NMM2 PING): TI realigns spike timing, not mean rate
 (black: E spikes; blue: population rate $r_E(t)$; red: TI envelope; boxed: envelope-locked timing, lock-in at Δf)

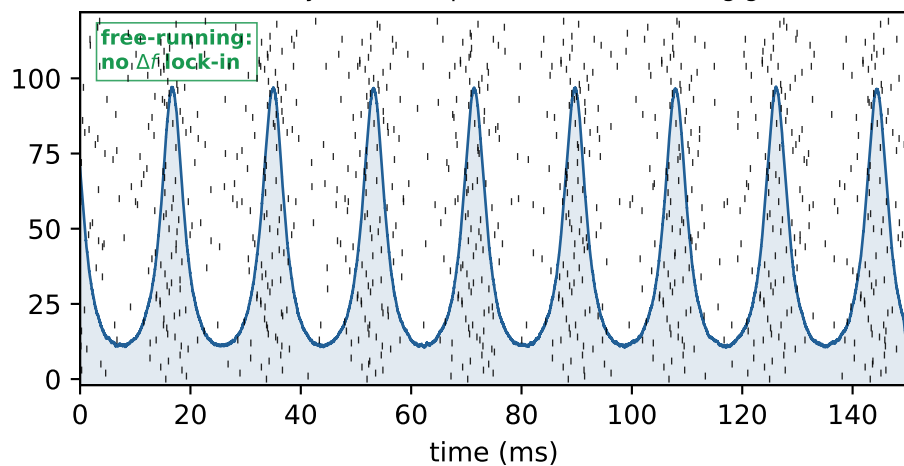
(a) Forced (below Hopf), TI OFF: asynchronous



(b) Forced, TI ON ($\Delta f=55$ Hz): timing bunched



(c) Oscillatory (above Hopf), TI OFF: free-running gamma



(d) Oscillatory, TI ON ($\Delta f=42$ Hz): TI beat gates the gamma (Δf -PAC)

